Sarah Barnes Haddix

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Education

The University of North Carolina at Chapel Hill — Chapel Hill, North Carolina, USA

B.S. Computer Science, B.S. Mathematics — 08/2022 - 05/2026

Graduated with Honors in Computer Science (completed and defended a senior honors thesis)
Graduated with Distinction (awarded for a cumulative GPA of 3.50 or higher)
Member, Honors Carolina (academic honors program admitting under 10% of each incoming class)
Dean's List: Fall 2022 – Spring 2026 (8 semesters)
Major GPA: 3.90/4.00 Overall GPA: 3.79/4.00

Professional Experience

The University of North Carolina at Chapel Hill — Chapel Hill, North Carolina, USA

Undergraduate Research Assistant — 04/2024 - 06/2026

- Worked with Computer Science Department Chair Dr. James H. Anderson's Real-Time Systems Research Group, focusing on real-time operating systems, AI, and GPU scheduling
- Assisted graduate students in conducting experiments, for example exploring GPU memory constraints and overload behaviors, and provided feedback and proofreading on research papers
- Co-authored a real-time robotics paper accepted at ICRA25; major contributions included the development and modification of software in Python and C++ to control a simulated drone leveraging Linux and ROS and the optimization of a path planning neural network using Tensorflow and ONNX

Owl Cyber Defense — Morrisville, North Carolina, USA

Engineering Intern — 06/2025 - 06/2026

- Worked in Rust and Java to develop cross-domain network security solutions for US government cybersecurity applications
- Added secure chat capabilities to existing products including the V2CDS, an NCDSMO Baseline Certified technology, using open source projects like Apache Nifi and OpenFire to handle streaming XML/XMPP data. Work on this project also included building a PostgreSQL-backed dashboard and management of Linux environments and virtual machines
- Worked on developing a secure RDP client/server in Rust for use in classified environments
- Held a U.S. Secret clearance

Axai Tech — Cape Town, Western Cape, South Africa

Project Intern — 06/2024

- Developed and applied deep learning models for protein sequence analysis and generation, supporting research into minimally invasive cancer therapies.
- Developed and trained multiple deep learning models, including LSTM-based architectures and transformer models, using PyTorch and Hugging Face
- Performed data preprocessing, cleaning, transformation, and exploratory analysis using Python, NumPy, and Pandas to prepare biological datasets for machine learning applications

Scale AI — Remote

Software Engineer for AI Training Data — 05/2024 - Present

- Contributing to coding-related AI training data in mainly Java, Python, and Swift used to finetune modern LLMs

Lenovo — Raleigh, North Carolina, USA

Intern — 05/2021 - 06/2021

- Collaborated with a team of three other interns to compile and present research on emerging technologies (NFTs, cryptocurrencies) to a Lenovo executive

Publications

- **"On the Necessity of Real-Time Principles in GPU-Driven Autonomous Robots,"** Syed W. Ali, Angelos Angelopoulos, Denver Massey, Sarah Haddix, Alexander Georgiev, Joseph Goh, Rohan Wagle, Prakash Sarathy, James H. Anderson, Ron Alterovitz, *in Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA)*, March 2025

Activities & Projects

- **A Proportional Share Resource Allocation Algorithm For Real-Time, Time-Shared Systems: A Closer Look**; honors thesis proving time bounds and correctness for the earliest eligible virtual deadline first (EEVDF) scheduling algorithm and analyzing its implementation in the Linux kernel, passed thesis defense with Honors
- **President of The UNC Chapter of the Association of Women in Mathematics**; organized and publicized events such as study nights, alumni/faculty panels, and social events to help foster a sense of community for women in the mathematics department; club is advised by Dr. Caroline Moosmueller
- **Selected for UNC's Undergraduate CyberSecurity Engagement Joint Program with The University of Nagoya in Japan**; exchange program including a trip to Nagoya, Japan to explore the intersection of cybersecurity, national security, and foreign policy through workshops, lectures, and site visits with industry and government leaders
- **Formal Verification of Onion Franking Security Properties Using Tamarin**; semester long project in COMP 590/637 to formally verify certain properties of a metadata-hiding cryptographic protocol that enables anonymous abuse reporting, proposed by researchers at The University of North Carolina in *Onion Franking: Abuse Reports for Mix-Based Private Messaging*, NDSS 2025
- **Member of UNC AI Club**; Developed a computer vision model for real-time wildfire detection optimized with TensorRT to be deployed on a Jetson-powered drone. Implemented a novel messaging protocol to stream live fire detection data from the drone's camera to a remote server
- **Awarded "Best Functionality" in Carolina App Team's Showcase, Spring 2024**, for a fully functional recreation of the Shazam app using ShazamKit API; completed App Team Carolina's 2 semester-long iOS development program, building several functional iOS apps